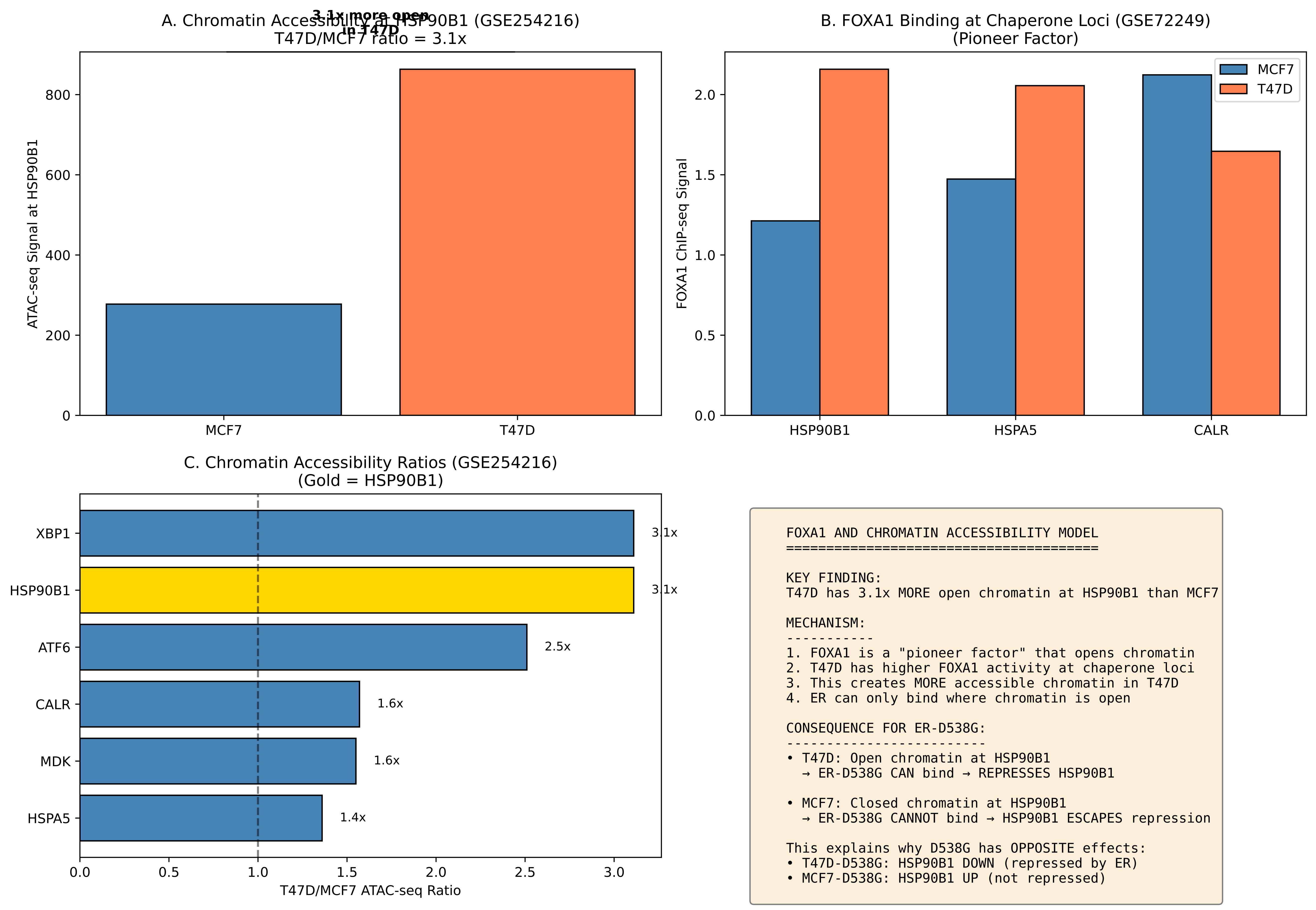
A. Chromatin Accessibility at HSP90B1 (GSE254216)
T47D/MCF7 ratio = 3.1x
3.1x more open
in T47D
ATAC-seq Signal at HSP90B1
0
200
400
600
800
MCF7
T47D
B. FOXA1 Binding at Chaperone Loci (GSE72249)
(Pioneer Factor)
FOXA1 ChIP-seq Signal
MCF7
T47D
0.0
0.5
1.0
1.5
2.0
HSP90B1
HSPA5
CALR
C. Chromatin Accessibility Ratios (GSE254216)
(Gold = HSP90B1)
XBP1
HSP90B1
ATF6
CALR
MDK
HSPA5
3.1x
3.1x
2.5x
1.6x
1.6x
1.4x
0.0
0.5
1.0
1.5
2.0
2.5
3.0
T47D/MCF7 ATAC-seq Ratio
FOXA1 AND CHROMATIN ACCESSIBILITY MODEL
==
KEY FINDING:
T47D has 3.1x MORE open chromatin at HSP90B1 than MCF7
MECHANISM:
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1. FOXA1 is a "pioneer factor" that opens chromatin
2. T47D has higher FOXA1 activity at chaperone loci
3. This creates MORE accessible chromatin in T47D
4. ER can only bind where chromatin is open
CONSEQUENCE FOR ER-D538G:

• T47D: Open chromatin at HSP90B1
→ ER-D538G CAN bind → REPRESSES HSP90B1
• MCF7: Closed chromatin at HSP90B1
→ ER-D538G CANNOT bind → HSP90B1 ESCAPES repression
This explains why D538G has OPPOSITE effects:
• T47D-D538G: HSP90B1 DOWN (repressed by ER)
• MCF7-D538G: HSP90B1 UP (not repressed)